Inverted MAML Card 1: Round 11 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±78 LSB

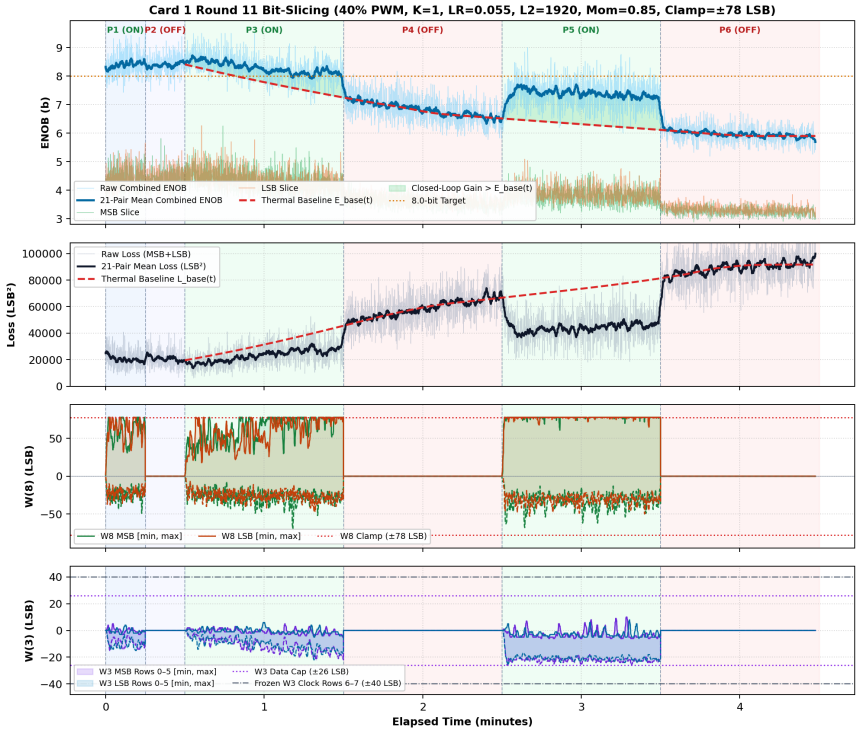
1. Round 11 Summary & Cumulative Comparison (R1–R10)

- **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).
- **Round 11 Detrended Trend Metrics:** Lift vs E_base = **+0.96 b**, Loss Ratio = **1.67x**, Climb = **+1.13 b**, Jitter = **0.40 b**, Recovery = **0.00**, Score = **+1.898**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R3dc_fix	0.02	2560	0.8	±70	+0.04b	1.00x	+0.24b	0.34b	1.00	+0.204
R4dc_fix	0.05	1792	0.85	±70	+1.08b	1.78x	+0.97b	0.40b	0.50	+2.480
R5dc_fix	0.055	1792	0.85	±85	+1.12b	1.77x	+1.37b	0.42b	0.00	+2.393
R6dc_fix	0.055	1792	0.8	±90	+0.99b	1.72x	+1.02b	0.40b	0.00	+1.770
R7dc_fix*	0.055	2048	0.85	±75	+1.04b	1.74x	+1.17b	0.40b	0.50	+2.665
R8dc_fix	0.055	2048	0.85	±75	+1.16b	1.85x	+0.68b	0.42b	0.00	+1.855
R9dc_fix	0.055	1920	0.85	±75	+0.81b	1.59x	+0.66b	0.38b	0.50	+1.756
R10dc_fix	0.055	1920	0.85	±78	+0.96b	1.67x	+1.13b	0.40b	0.00	+1.898

2. Phase-by-Phase Telemetry (4,181 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	322	8.18->8.49 (8.40b)	-	26056->18707	[78,78]	15	322
P2	Cold / Regler OFF	290	8.26->8.42 (8.38b)	-	24582->19959	[0,0]	0	0
P3	Warm / Regler ON	1040	8.50->8.10 (8.29b)	+0.66 b	17497->27274	[78,78]	22	1040
P4	Warm / Regler OFF	935	7.37->6.60 (6.85b)	+0.01 b	43838->64765	[0,0]	0	0
P5	Warm / Regler ON	836	6.78->7.30 (7.38b)	+1.12 b	59041->46169	[78,78]	26	836
P6	Warm / Regler OFF	758	6.19->5.89 (5.97b)	-0.01 b	79196->92490	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 11

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±78 LSB, while W(3) rows 0–5 stay within ±26 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 11 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

